

Appendix 6 - Cross-Model Consistency Evaluation

As mentioned in Section 3.3, this appendix reports the results of the consistency evaluation replicated across two additional large language models, Claude Sonnet 4-6 and DeepSeek-Chat, alongside GPT-4.1. The same three metrics were applied to 20 independent runs per room per model (180 runs in total).

Table A1. Role Escape Rate across Three Models

Room	GPT-4.1	Claude Sonnet 4-6	DeepSeek-Chat
Researcher vs. Red Guard Room	0.0%	0.0%	0.0%
Intergenerational Understanding Room	0.0%	0.0%	0.0%
User vs. Sent-Down Youth Q&A Room	0.0%	0.0%	0.0%

Note. Rate = proportion of agent turns containing content prohibited by the prompt specification. User turns excluded.

Table A2. Cross-Run Semantic Similarity across Three Models

Room	GPT-4.1 Mean (SD)	Claude Sonnet 4-6 Mean (SD)	DeepSeek-Chat Mean (SD)
Researcher vs. Red Guard Room	0.73 (0.02)	0.66 (0.08)	0.46 (0.08)
Intergenerational Understanding Room	0.80 (0.07)	0.76 (0.08)	0.66 (0.14)
User vs. Sent-Down Youth Q&A Room	0.58 (0.11)	0.52 (0.07)	0.42 (0.13)

Note. Mean = average pairwise cosine similarity across agent slots (190 pairs per slot). Computed using all-mpnet-base-v2 (Reimers and Gurevych 3982). User and Moderator roles excluded.

Table A3. LSA-Based Run-Level Thematic Consistency across Three Models

Room	GPT-4.1 Mean (SD)	Claude Sonnet 4-6 Mean (SD)	DeepSeek-Chat Mean (SD)
Researcher vs. Red Guard Room	0.35 (0.04)	0.31 (0.04)	0.19 (0.07)
Intergenerational Understanding Room	0.54 (0.09)	0.46 (0.10)	0.47 (0.10)

User vs. Sent-Down Youth Q&A Room	0.20 (0.04)	0.29 (0.05)	0.19 (0.05)
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Note. Mean = average pairwise cosine similarity between run-level LSA vectors within each room (N = 20 runs per room, 190 pairs).

Figure A1. LSA-Based Topic Clustering of 60 Simulation Runs (Claude Sonnet 4-6)

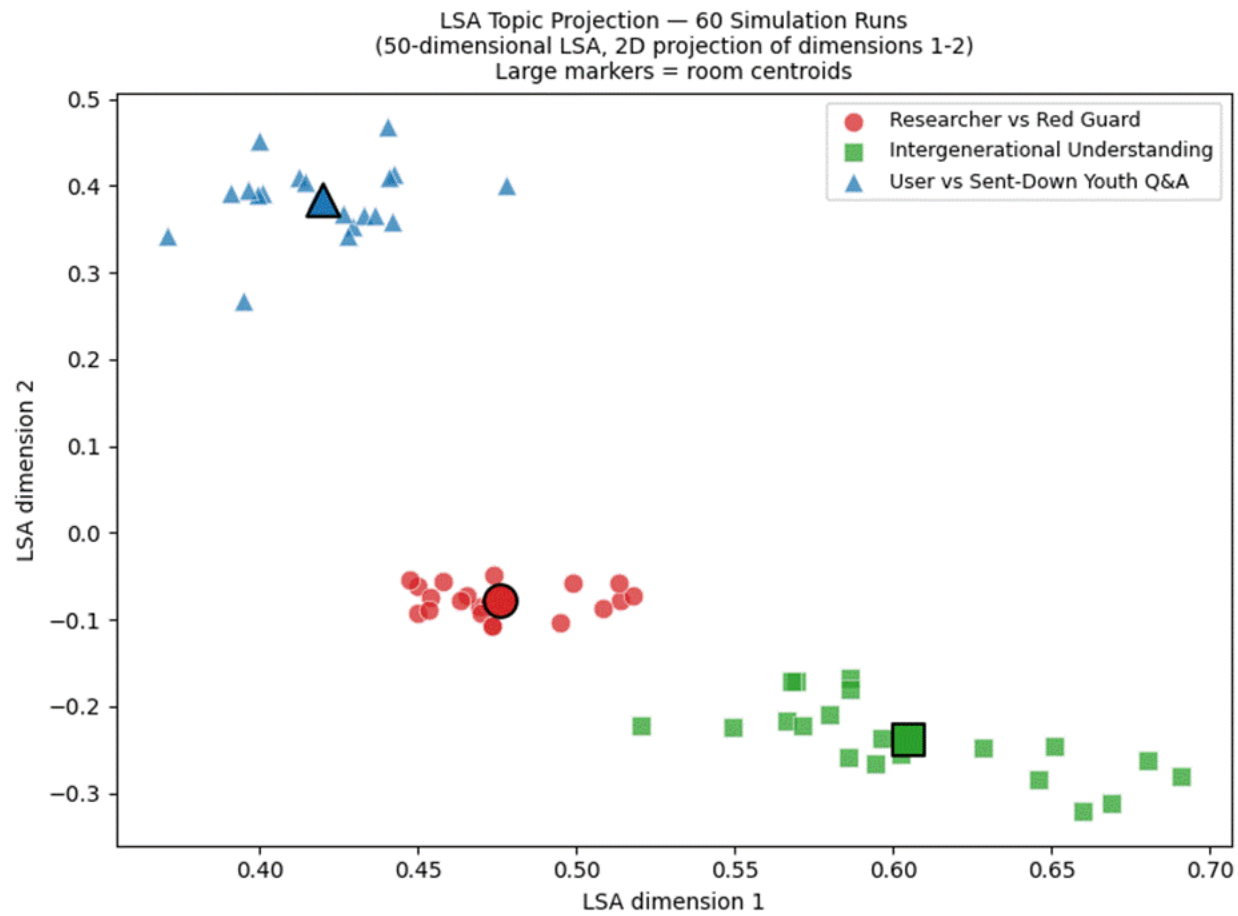


Figure A2. LSA-Based Topic Clustering of 60 Simulation Runs (DeepSeek-Chat)